

Logistic Regression Binary dataset:

<https://www.kaggle.com/datasets/mathchi/diabetes-data-set?resource=download>

The dataset contains medical information from 768 female patients, each described by eight health measurements, including blood sugar levels, BMI, age, and number of pregnancies. Every patient also has a recorded diabetes diagnosis: 0 means no diabetes, 1 means diabetes. Think of it as a medical records table where each row is one patient and each column tells us something about her health. We want to teach a computer to look at a new patient's health numbers and predict whether she is likely to have diabetes.

There are only two possible answers yes or no which is why this is called a binary classification problem. The tool we used was Logistic Regression. It assigns a weight to each health measurement based on how strongly it relates to diabetes, then combines them into a probability score. Before training, we split the data into a training set (75% of patients, used to teach the model) and a test set (the remaining 25%, used to check how well it learned). We also applied feature scaling so that large values like glucose don't accidentally overshadow smaller ones like a pedigree score. Once prepared, the model learned relationships between the health measurements and diabetes outcomes from the training data and learned which combinations of health factors most reliably indicate diabetes.

Testing on patients the model had never seen before showed strong performance across accuracy, precision, recall, and AUC score. Of these, recall matters most in a medical context because missing a real diabetes case is far more costly than raising a false alarm. The model independently identified Glucose as the strongest predictor, followed by BMI and Age, exactly the risk factors doctors already rely on. When a machine learning model rediscovers what medical professionals know, it is a strong sign the model is working correctly and not just finding random patterns. The final output is a live tool: enter any patient's eight health values and the model instantly returns a prediction, "No Diabetes Detected" or "Diabetes Detected" along with the actual probability behind that decision.

Logistic Regression Multinomial dataset:

<https://www.kaggle.com/datasets/muratkokludataset/dry-bean-dataset>

The Dry Bean Dataset contains measurements taken from photographs of 13,611 dry beans across seven varieties: Seker, Barbunya, Bombay, Cali, Dermason, Horoz, and Sira. Instead of measuring beans by hand, researchers used image processing software to automatically extract 16 shape-based measurements from each photo, including features such as length, roundness, compactness, and area. Think of it like a digital ruler that captures every curve and edge of a bean from a picture. The technique we used was Multinomial Logistic Regression. Our model studies the training data and learns a set of mathematical weights that connect the bean measurements to each bean variety. When the model sees a new bean, it calculates a probability for each variety, something like "82% chance this is Dermason, 7% Cali, 5% Seker," and then selects the class with the highest probability as its prediction.

After training on roughly 9,500 beans and testing on about 4,000 beans it had never seen before, the model correctly identified the variety 92.2% of the time, meaning it would correctly sort about 92 out of every 100 beans. It performed perfectly on Bombay beans, which

are noticeably larger and more distinct. The only real difficulty occurred between Dermason and Sira, two varieties with very similar shapes, much like how a non-expert might confuse two similar apple varieties at a glance. The practical value of this approach is clear. A food processing facility could place a camera above a conveyor belt and use this model to automatically sort beans by variety, faster and more consistently than any manual process.

Regression Multivariate dataset:

<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

The bike sharing dataset represents data for a bike sharing system where users can rent a bike from one place and return to another place. There were a total of 731 datapoints representing daily bikes rented everyday over two years, from 2011 to 2012. The target variable of the data is the daily count of bikes rented and includes corresponding daily variables that might contribute to daily bikes rented. The dataset has 17 total variables, out of which 10 were chosen for regression analysis. The variables chosen are temperature, weekday (day of the week), season, year, windspeed, humidity, holiday, workingday, and weathersit, a variable representing the description of weather. These variables are picked due to how they may impact and play a role in the daily rented bikes for a given day. This is a multivariate regression analysis since there are multiple independent variables chosen to analyze the relationship between the number of bikes rented daily and corresponding daily statistics and factors that may contribute to it.

For regression analysis, dataset features were defined to the 10 chosen variables and target set to the daily count of bikes rented. Next a model was trained and created to enable predictions based on the dataset. The model has a coefficient of determination, R^2 , of 0.84, this means 84% of the variance in daily number of bikes rented is explained by the model based on the selected predicting variables. This is a strong model, and the features selected were a good fit for explaining the daily bikes rented. The model has a mean squared error (MSE) of about 634351, this helps explain how close points are to the predicted values, the smaller the number the better. Altogether proportionally, it represents a good average error out of the daily count possible in this case.

The scatter plot of actual versus predicted values shows how close the predicted and actual points lie on the regression model. This means the model has strong performance, where predictions are very close to the daily actual bikes rented. Overall, the multivariate regression model developed for the bike sharing system can be used to predict the number of daily bikes for years 2011 and 2012 based on selected factors. In other words, this model provides good predictions based on 2011 and 2012 trends and factors.

Regression Linear dataset:

<https://archive.ics.uci.edu/dataset/597/productivity+prediction+of+garment+employees>

The garment dataset involves productivity of garment employees and factors that might impact it. This dataset contains two departments: sewing and finishing of which each team is a part of. For this regression analysis, just the sewing department was chosen and filtered for. After filtering for the sewing department, there were a total of 691 datapoints. After analyzing the variables, the incentive variables for predicting and modeling productivity were chosen. The incentive variable stood out as a strong factor, the rest of the variables seemed to show much weaker relationships. During trial-and-error modeling, the other variables did not make much of an impact on the model. Therefore, out of the 14 factors, for this analysis just the incentive factor was chosen. The incentive variable here represents a financial bonus in Bangladeshi Taka (BDT). This model is considered a simple linear regression, since one variable is being used to estimate the productivity outcome for the garment workers in the sewing department.

The regression model was trained and created on the selected feature (predicting variable) and target, which was the productivity of the workers. The model has a coefficient of determination, R^2 , of 0.65 this means 65% of the productivity of workers for the sewing department is explained by the incentive features. This is a strong model given the dataset. The mean squared error (MSE) is 0.007, this is low and close to 0 which indicates how the model is very close to the actual productivity of workers.

Visualizing the actual versus predicted productivity for the sewing department on the scatterplot shows how closely the points align with the model. Some points show an exact match, others are close, while there are some outliers as well, however the model is a strong fit for data overall. Overall, the model is a strong fit and shows incentive plays a major role in predicting the productivity of garment workers for the sewing department. This prediction model may be valuable in the garment industry when it comes to maximizing productivity of garment workers pertaining to the sewing department.